BASICS OF NETWORKS & NETWORK TOPOLOGIES

Topics to be covered

Module 3.2 from the syllabus : Network topologies: Star, Mesh, Ring, Hybrid

Networks

i. **Three network criteria**
ii. **Network physical structure**
    a. **Type of connection**
    b. **Physical topology**
        1. **Mesh**
        2. **Star**
        3. **Bus**
        4. **Ring**

NETWORKS

- A **set of devices** (or ***nodes***) connected by **communication links**.
- A **node** can be a computer, printer, or any other device capable of sending data and/or receiving data generated by other nodes on the network.
- A **link** is a communications pathway that transfers data from one device to another.

Three Network Criteria

1. **Performance**
2. **Reliability**
3. **Security**

Three Network Criteria

1. **Performance**
   - 2 important ways are to measure performance :
     1. **Transit time** is the amount of time required for a message to travel from one device to another.
     2. **Response time** is the elapsed time between an inquiry and a response.
   - The performance of a network depends on factors such as –
     - the **number of users**,
     - the **type of transmission medium**,
     - the **capabilities of the connected hardware**, and
     - the **efficiency of the software**.
   - 2 **Networking Metrics** to **evaluate** performance are **throughput** and **delay**.
   - We often need more throughput and less delay. However, these two criteria are often contradictory. If we try to send more data to the network, throughput increase but the delay increase too because of traffic congestion in the network.

Three Network Criteria

2. **Reliability**
   - In addition to accuracy of delivery, network reliability is measured by :
     - the **frequency** of failure
     - the **time it takes for a link to recover from a failure**, and
     - the network's **robustness in a catastrophe**.

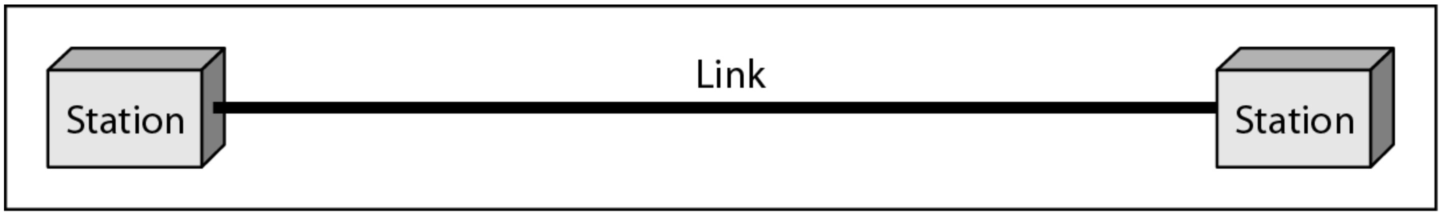
Three Network Criteria

3. **Security**
   - Network security issues include :
     - **protecting** data from **unauthorized access**,
     - **protecting** data from **damage and development**, and
     - **implementing policies and procedures** for recovery from breaches and data losses.

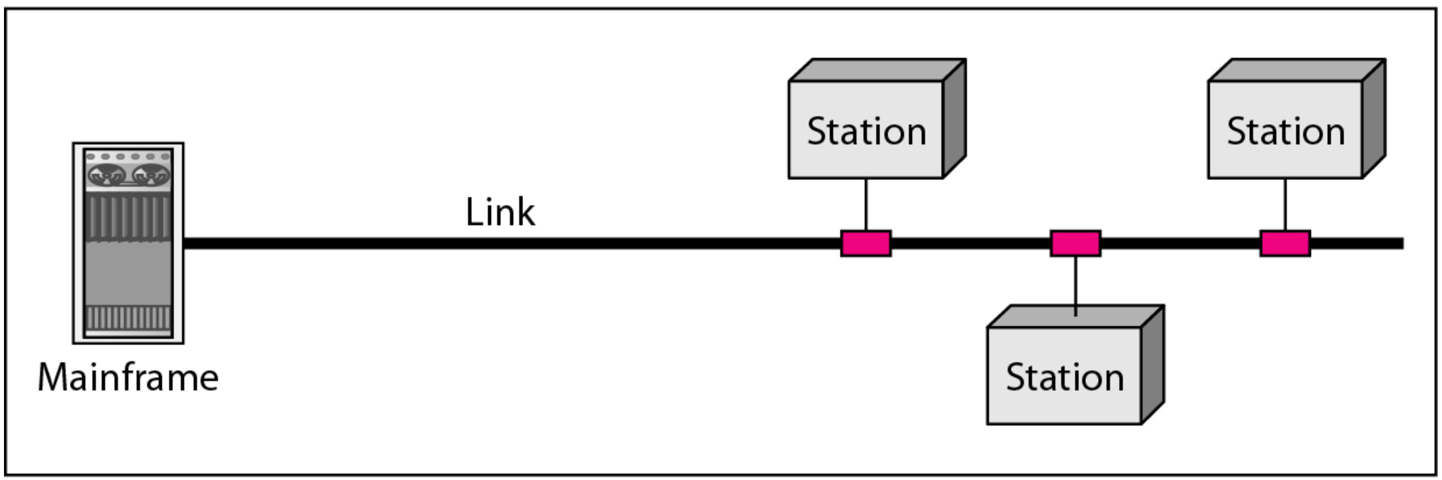
Physical Structures -
Type of Connection

- **Point-to-Point** : a dedicated link between two devices.
  - Link options : wire or cable to connect the two ends or microwave or satellite links.
- **Multipoint** (also called **multidrop**): more than two specific devices share a single link.
  - Here the **capacity of the channel is shared**, either spatially or temporally.
  - If several devices can use the link simultaneously, it is a ***spatially shared*** connection.
  - If users must take turns, it is a ***timeshared*** *connection.*

Figure 1.3 *Types of connections: point-to-point and multipoint*



a. Point-to-point

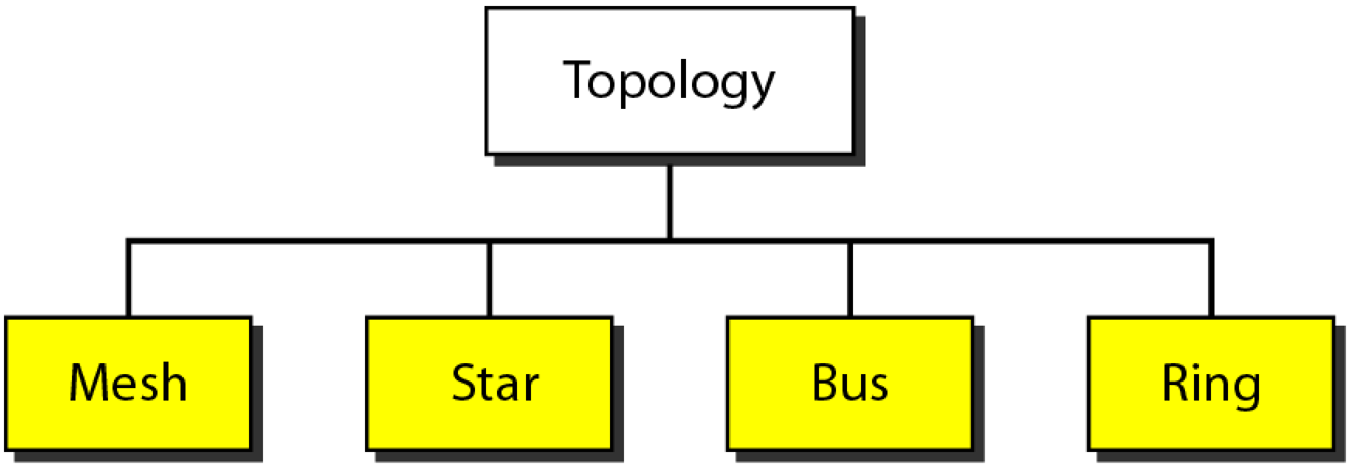


b. Multipoint

Physical Structures –
Physical Topology

- Refers to the way in which a **network is laid out physically** OR the **geometric representation** of the relationship of all the links and linking devices (usually called nodes) to one another.
- Four basic topologies possible:
  1. mesh
  2. star
  3. bus
  4. ring

Figure 1.4 *Categories of topology*

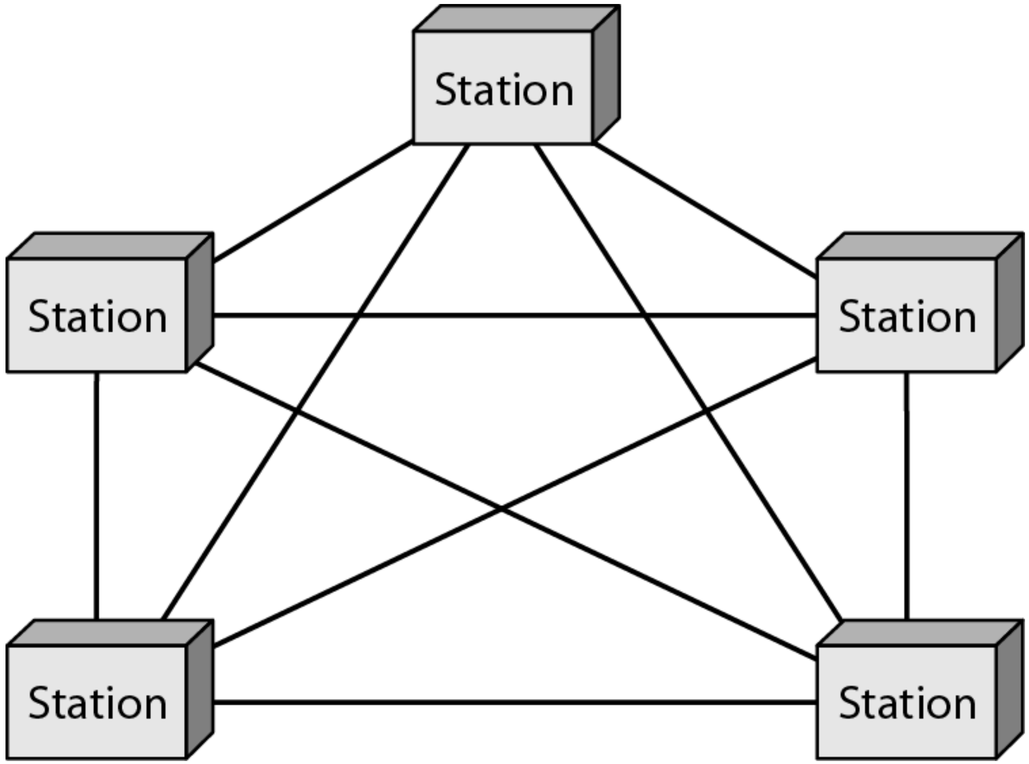


Mesh topology

- Every device has a **dedicated point-to-point link** to every other device.
- In a fully connected mesh network with ***n*** *nodes,* we need ***n(n - 1) simplex physical* links**.
- If each physical link allows duplex mode, we need ***n(n -1) /2* duplex-mode links**.
- i/e., every device on the network must have ***n - 1***

  I/O ports *(see Figure 1.5) to be connected to the other (n – 1) stations.*

Figure 1.5 *A fully connected mesh topology (five devices)*

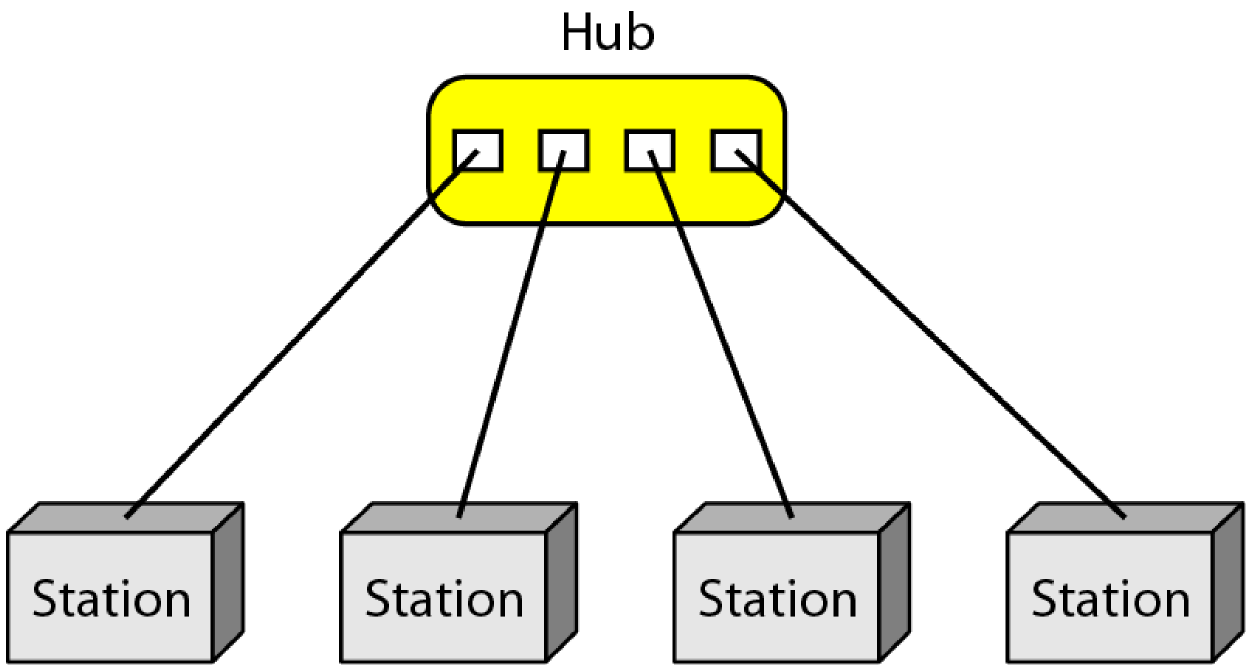


Advantages & disadvantages of Mesh Topology

- **Advantages**
  - Dedicated links eliminate the traffic problems
  - **Robust**: one unusable link does not incapacitate the entire system
  - **Privacy** or **security**
  - Easy **fault identification** & **fault Isolation :** Traffic can be routed to avoid links with suspected problems
- **Disadvantages**
  - Large number I/O ports & **cabling** required
  - Installation & reconnection are **difficult**
  - Wiring can be greater than **available space**
  - Hardware required to connect each link (I/O ports & cable) can be **very expensive**

Example: the connection of **telephone regional offices** in which each regional office needs to be connected to every other regional office.

Figure 1.6 *A star topology connecting four stations*



Star Topology

- Each device has a dedicated point-to-point link **only to a central controller** (**hub**)
- No direct traffic between devices. The controller acts as an **exchange**.

Advantages & disadvantages of Star Topology

Advantages

- **Less expensive** than mesh topology
- **Installation** & **reconfiguration** are **easy**
- Required **less cabling** than mesh topology
- **Robust**: If one link fails, only that link is affected
- **Easy fault identification** & fault **Isolation**

Disadvantages

- **Dependency** of whole topology **on single point**

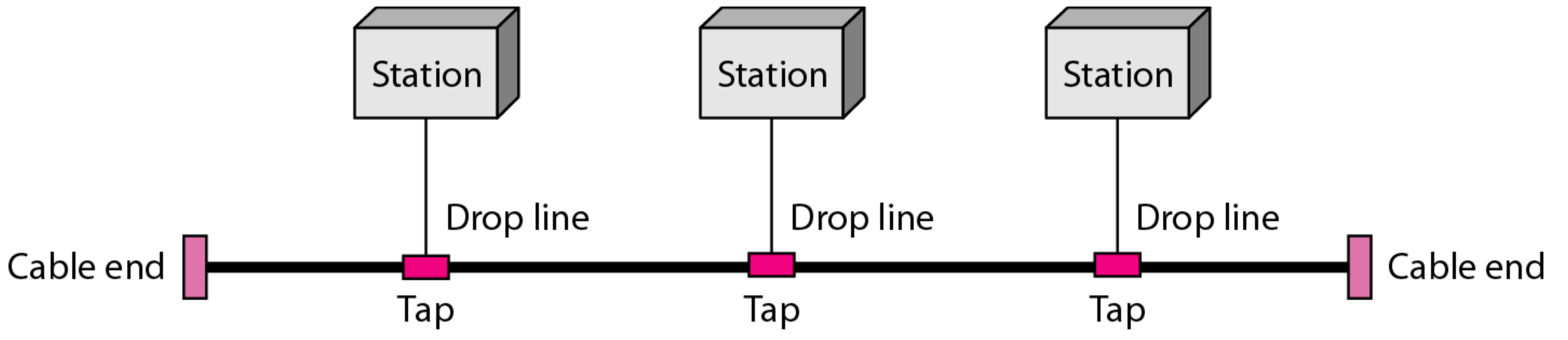
Application: High Speed LAN

Example: used in local-area networks (LANs)

Bus Topology

- One **long cable** acts as a **backbone to link all** the devices in a network (see Figure 1.7)
- Nodes are connected to the bus cable by **drop lines** and **taps**.
- As a signal travels along the backbone, some of its energy is transformed into heat making it weaker. For this reason there is a **limit on the number of taps** and on **the distance between those taps**.

Figure 1.7 *A bus topology connecting three stations*



Advantages & disadvantages of Bus Topology

Advantages

- **Ease of installation**
- Requires **less cabling**: Backbone cable can be laid along the most efficient path, then connected to the nodes by drop lines of various lengths

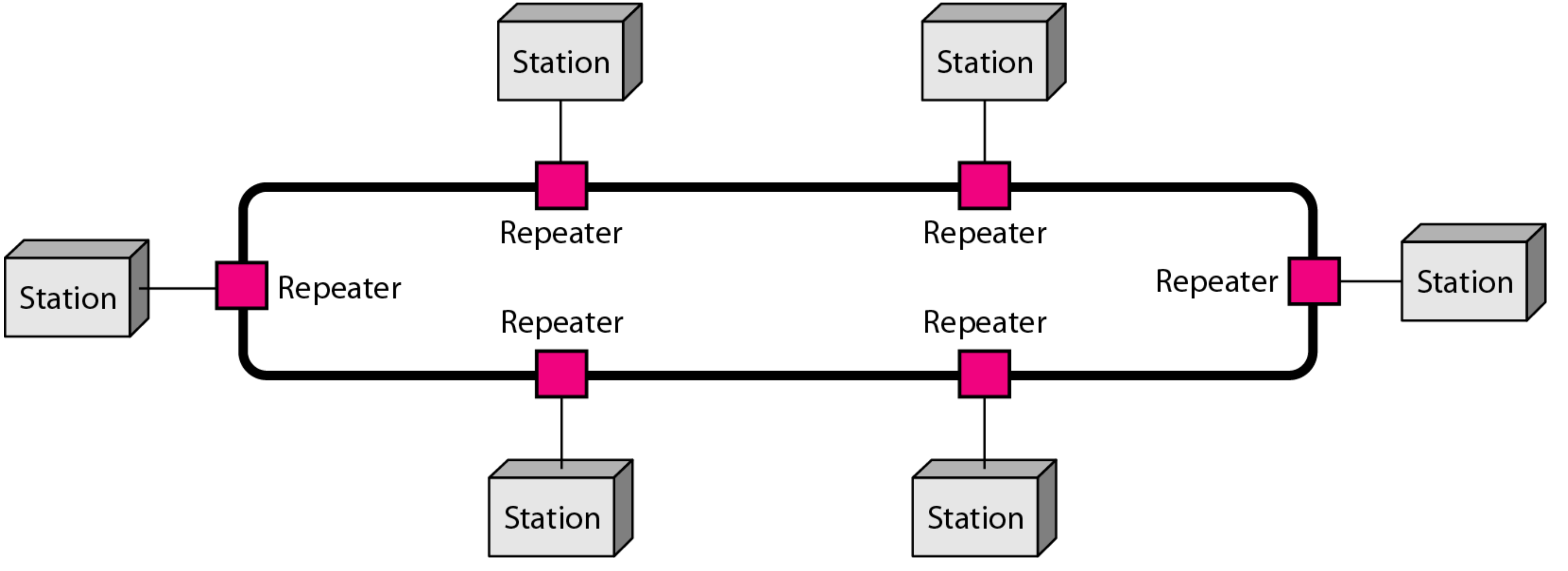
Disadvantages

- **Difficult reconnection & fault isolation**
  - bus is usually designed to be optimally efficient at installation. Hence, difficult to add new devices
- **Signal reflection at the taps** can cause degradation in quality
- A fault or break in the bus cable **stops all transmission**

Ring Topology

- Each device has a dedicated point-to-point connection with only the two devices on either side of it
- A signal is passed along the ring in one direction, from device to device, until it reaches its destination
- Each device incorporates a **repeater**: repeater regenerates the bits and passes them along when a device receives a signal intended for another device

Figure 1.8 *A ring topology connecting six stations*



Advantages & disadvantages of Ring Topology

- **Advantages**
  - **Easy to install** & **reconfigure**
  - **fault isolation** is **simplified**
- **Disadvantages**
  - **Unidirectional traffic** : a break in ring (such as a disabled station) can disable the entire network.

Hybrid Topology

- Combining different topologies

Figure 1.9 *A hybrid topology: a star backbone with three bus networks*